

Foundation (Jalapeno)

- Q1.** Simplify $\sqrt{8} + \sqrt{8}$
(A) 4
(B) $2\sqrt{2}$
(C) 8
(D) $4\sqrt{2}$
(E) 16
- Q2.** Simplify $\sqrt{50} - \sqrt{32}$
(A) $\sqrt{18}$
(B) $\sqrt{2}$
(C) $5\sqrt{2} - 4\sqrt{2}$
(D) 4
(E) $\sqrt{2}$
- Q3.** A stock's value increases by $\sqrt{20}\%$. If the initial value is \$100, what is the new value?
(A) \$106
(B) \$110
(C) \$120
(D) \$122
(E) \$140
- Q4.** Simplify $(\sqrt{3} \cdot \sqrt{12})^2$
(A) 36
(B) 27
(C) 48
(D) 12
(E) 9
- Q5.** Simplify $\frac{\sqrt{72}}{\sqrt{8}}$
- (A) $\frac{3\sqrt{2}}{2}$
(B) $\frac{3\sqrt{3}}{2}$
(C) 3
(D) 6
(E) $3\sqrt{2}$
- Q6.** A company has \$1000 to invest. They split the investment into two parts: one part grows at $\sqrt{10}\%$, the other at $\sqrt{20}\%$. What is the total growth?
(A) \$1005.47
(B) \$1100
(C) \$1050
(D) \$1200
(E) \$1150
- Q7.** Simplify $\sqrt{48} - \sqrt{18}$
(A) $\sqrt{10}$
(B) $2\sqrt{6}$
(C) $4\sqrt{3}$
(D) $2\sqrt{2}$
(E) $\sqrt{30}$
- Q8.** A \$1000 investment grows by $\sqrt{50}\%$ in the first year. What is the value after two years, assuming the same growth rate?
(A) \$1154.70
(B) \$1500
(C) \$1600
(D) \$1200
(E) \$1225

Open Ended Questions (FRQ)

- Q9.** Simplify $\frac{\sqrt{200}}{\sqrt{8}}$ and explain the reasoning behind your answer

- Q10.** A company invests \$5000, *split into two parts*: one grows at $\sqrt{15}\%$, the other at $\sqrt{30}\%$. If the first part grows for 3 years, and the second part for 2 years, calculate the total value after the given periods, assuming compound growth

Chef's Tips

- Tip 1: Ensure students simplify radicals before performing operations
- Tip 2: Use examples with real-life applications, like finance and business, to increase engagement
- Tip 3: Provide step-by-step guidance on complex problems, focusing on explaining each operation with radicals